Optimisation and Deep Learning

Test Specification Table

CLO1	Justify neural network algorithms for data analytic applications in real-world problems (A3, PLO9)	Assignment (Report)
CLO2	Apply deep learning methods with Optimisation concepts for practical solutions (C3, PLO2)	

Assignment 1 (40%):

Question No.		Question Vs Taxonomy Affective Level						
	Topic	1	2	3	4	5		PLO
		SQ	SQ	SQ	SQ	SQ		
1	Data Preparation and Pre-processing, Artificial Neural Network					100 %		9
	Total					**100 %**		

Assignment 2 (60%):

Question No.		Question Vs Taxonomy Cognitive Level						
	Topic	1	2	3	4	5	6	PLO
		SQ	SQ	SQ	SQ	SQ	SQ	
1	Application of Deep Learning			100 %				2
	Total			**100 %**				

Marking Scheme

Assignment 1 (40 marks)	Weightage	Marks
Task 1: Dataset & Algorithm selection	**20**	
Task 2: Data Preparation	**20**	
Total		
Comments:		

Assignment 2 (60 marks)	Weightage	Marks
Task 1: Model Building	**30**	
Task 2: Model Tuning	**10**	
Task 3: Model Evaluation & Discussion	**10**	
Task 4: Conclusion	**10**	
Total		
Comments:		

Final marks & grade

Assignment 1 (40 marks)	
Assignment 2 (60 marks)	
Total	
Grade	

Marking Criteria

The following guidelines indicate the standard that will be expected for each grade.

Distinction (75% and above)

Demonstrates an excellent understanding of the overall objective of the project and purpose of optimization in predictive modeling. The report features excellent critical analysis for each of the project steps. Excellent presentation of work with details of effort and ability to explain the way in which he/she has approached the problem. The proposed solution exceeds expectation by going beyond in including additional insights or methodologies that are the result of student's self-learning.

Merit (65 – 74%)

Demonstrates a good understanding of the problem. Good discussion on the selected tools, methodologies, and content. The report features, almost all learning outcome with relevant discussion. All the codes are working and producing results. Documentation will be satisfactory with suitable illustrations. May have minor critical analysis/discussion. The student can discuss his/her idea at the presentation and is able to confidently explain the approach adapted and the results.

Pass (50 – 64%)

Demonstrates some understanding of the problem. The proposed solution is implementable although with some errors and/or omissions. The assumptions for the proposed solution with relevant strategies are logical but with some errors and/or omissions Documentation is adequate. The student can discuss his/her implementation at the presentation and is able to explain the way in which he/she has attempted the project. The report contains reasonably good presentation of the results and analysis.

Fail (0 - 49%)

Demonstrates poor understanding of the problem / does not understand the problem. The steps in the project are implemented solution partially / not implemented. The proposed solution has limited discussion and major errors and/or omissions. Documentation is incomplete. Poor explanation during demo.

RUBRICS	5 Distinction	4 Credit	3 Pass	2 Marginal Fail	1 Fail	0 Zero
Assignment 1 (40 marks)						
Task 1 **20%**	Clear, complete and detail description explanation and justification of the dataset & algorithms selection and complete literature review. All relevant information are provided.	Clear, complete yet very basic description, explanation and justification of the dataset & algorithms selection and literature review are provided but briefly discussed/identified. No irrelevant information are found.	Brief/vague description, explanation and justification of the dataset & algorithms selection and literature review. Some irrelevant information is found.	Done but missing (one component) either the dataset& algorithms, justification of the selection and literature review. Mostly irrelevant information are provided.	Done but missing (two components) either the dataset& algorithms, justification of the selection and literature review. Mostly irrelevant information are provided.	No evidence in the documentation.
Task 2 **20%**	Clear and complete description of Exploratory Data Analysis (EDA), and data preparation including pre-processing. All relevant information is provided.	Clear, complete yet very basic description of Exploratory Data Analysis (EDA), and data preparation including pre-processing made. Briefly discussed/identified. No irrelevant information is found.	Brief/vague description of Exploratory Data Analysis (EDA), and data preparation including pre-processing. Some irrelevant information is found.	Done but missing (several components) either description of Exploratory Data Analysis (EDA), or data preparation including pre-processing. Mostly irrelevant information is provided.	Done but missing (most of the components) either description of Exploratory Data Analysis (EDA), or data preparation including pre-processing. Mostly irrelevant information is provided.	No evidence in the documentation.
Assignment 2 (60 marks)						
Task 1 **30%**	Clear, complete and detailed description of model initialization, training, and testing. No errors in the process. All relevant information is provided.	Clear, complete yet very basic description and explanation of model initialization, training, and testing but briefly discussed/identified. No irrelevant information are found.	Brief description and explanation of model initialization, training, and testing. Minor errors found in the process. Some irrelevant information is found.	Done but missing several aspects in the model initialization, training, and testing. Minor errors found in the process. Mostly irrelevant information is provided.	Done but missing most aspects in the model initialization, training, and testing. Major errors found in the process. Mostly irrelevant information is provided.	No evidence in the documentation.

Task 2 **10%**	Clear, complete, and detailed description & correct implementation of Model tuning. All the necessary information is included in the report.	Clear, complete yet very basic description & implementation of Model tuning is made but briefly discussed/identified with minor errors. All the necessary information is included in the report. No irrelevant information is found.	Brief description & implementation of Model tuning is made with minor errors. Minor errors in the information provided. Some irrelevant information is found.	Done but missing several description & implementation with major errors in the Model tuning. Missing some of the information. Mostly irrelevant information is provided.	Done but missing most of the description & implementation with major errors in the Model tuning. Missing some of the information. Mostly irrelevant information is provided.	No evidence in the documentation.
Task 3 **10%**	Clear, complete, and detailed description of Model Evaluation & Discussion is provided. Complete answer with all relevant information is found in the report.	Clear, complete yet very basic description of Model Evaluation & Discussion is provided but not precise. No irrelevant information is found.	Brief description Model Evaluation & Discussion is made with minor errors. Some irrelevant information is found.	The Model Evaluation & Discussion is made with minor errors. Incomplete and mostly irrelevant information are provided.	The Model Evaluation & Discussion is made with major errors. Incomplete and mostly irrelevant information are provided.	No evidence in the documentation.
Task 4 **10%**	Clear, complete, and detailed description of Final Evaluation & Conclusion is provided. Complete answer with all relevant information is found in the report.	Clear, complete yet very basic description of Final Evaluation & Conclusion is provided but not precise. No irrelevant information is found.	Brief description Final Evaluation & Conclusion is made with minor errors. Some irrelevant information is found.	The Final Evaluation & Conclusion is made with minor errors. Incomplete and mostly irrelevant information are provided.	The Final Evaluation & Conclusion is made with major errors. Incomplete and mostly irrelevant information are provided.	No evidence in the documentation.